

Tribhuvan University
Institute of Science and Technology

2077



Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology (PHY. 113)
(Physics)
(New Course)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.
The questions are of equal value.

Group A

Long Answer Questions:

Attempt any TWO questions

(2x10=20)

1. Explain RTL and TTL gates. How memory and clock circuits can be made by using these gates? Explain how they work?
2. Set up differential equation for an oscillation of a spring using Hooke's and Newton's second law. Find the general solution of this equation and hence the expressions for period, velocity and acceleration of oscillation.
3. Describe Frank-Hertz experiment. Interpret how the results of this experiment advocate atomic model proposed by Bohr?

Group B

Short Answer Questions:

(8x5=40)

Attempt any EIGHT questions:

4. Discuss magnetic dipole moment. What is its effect on atom? and on molecules? Explain.
5. Describe the following process of IC production: (a) Oxidation, (b) Pattern definition, and (c) Doping.
6. Explain the construction and working of bipolar junction transistor (BJT).
7. A roulette wheel with moment of inertia $I = 0.5 \text{ kg m}^2$ rotating initially at 2 rev/sec coasts to a stop from the constant friction torque of the bearing. If the torque is 0.4 Nm, how long does it take to stop?
8. Two large parallel plates are separated by a distance of 5 cm. The plates have equal but opposite charges that create an electric field in the region between the plates. An α particle ($q=3.2 \times 10^{-19} \text{ C}$, $m= 6.68 \times 10^{-27} \text{ kg}$) is released from the positively charged plate, and it strikes

the negatively charged plate 2×10^{-6} sec. later. Assuming that the electric field between the plates is uniform and perpendicular to the plates, what is the strength of the electric field?

9. In neutron spectroscopy a beam of monoenergetic neutrons is obtained by reflecting reactor neutrons from a beryllium crystal. If the separation between the atomic planes of the beryllium crystal is $0.732 \text{ \AA}$, what is the angle between the incident neutron beam and the atomic planes that will yield a monochromatic beam of neutrons of wavelength $0.1 \text{ \AA}$?

10. What is the probability of finding a particle in a well of width a at a position $a/4$ from the wall if $n = 1$, if $n = 2$, if $n = 3$. Use the normalized wavefunction $\psi(x, t) = \left(\frac{2}{a}\right)^{\frac{1}{2}} \sin\left(\frac{n\pi x}{a}\right) e^{-\frac{iEt}{\hbar}}$.

11. The energy gap in silicon is 1.1 eV , whereas in diamond it is 6 eV . What conclusion can you draw about the transparency of the two materials to visible light ($4000 \text{ \AA}$ to $7000 \text{ \AA}$)?

12. Find the truth table for the circuit shown in the figure. What logic function will the circuit perform if the constant $+5 \text{ V}$ input to the first two gates is changed to ground potential?

